

Material Handling

Objectives of Material Handling

Unit Load Concept

Principles of Material Handling

Types of Material Handling Equipment

Degree of Mechanization in Material handling

Definition

Material handling is the function of moving the right material to the right place in the right time, in the right amount, in sequence, and in the right condition to minimize production cost.

Objectives of Material Handling

1. To increase efficiency of material flow by ensuring availability of materials when & where they needed
2. To reduce Material Handling cost
3. To improve facilities utilization
4. To improve safety & working conditions
5. To facilitate manufacturing processes
6. To increase productivity

UNIT LOAD Concept

- It is defined as number of items arranged such that they can be handled as a single object
- Unit load can be accomplished by:
 - 1. Palletization:** It is assembling & Securing of individual items on a platform that can be moved by a truck or a crane



UNIT LOAD Concept (contd.)

2. Unitization: It is also the assembling of goods, but as one compact load. Unlike Palletization additional materials are used for packaging & wrapping the items as a complete unit



UNIT LOAD Concept (contd.)

3. Containerization: It is assembling of items in a box or a bin. It is most suitable for use with conveyors, especially for small items



Principles of Material Handling

1. **Planning:** Plan all MH & Storage activities to obtain maximum overall operating efficiency
2. **System Flow:** Integrate as many handling activities as is practical into a coordinated system of operations covering vendor, receiving, storage, production, inspection, packaging, warehousing, shipping, transportation, & customer
3. **Material Flow:** Provide an operation sequence & equipment layout optimizing material flow

Principles of Material Handling (Contd.)

- 4. Simplification:** Simplify handling by reducing, eliminating, or combining unnecessary movements &/or equipment
- 5. Gravity:** Use gravity to material handling whenever practical
- 6. Space Utilization:** Make optimum utilization of the building cube
- 7. Unit Size:** Increase the quantity, size or weight of unit loads or flow rate
- 8. Mechanization:** Mechanize handling operations
- 9. Automation:** Provide automation to include production, handling & storage functions

Principles of Material Handling (Contd.)

- 10. Equipment Selection:** In selecting handling equipment, consider all aspects of the material handled, the movement, & the method to be used
- 11. Standardization:** Standardize handling methods as well as type & sizes of handling equipment
- 12. Adaptability:** Use methods & equipment that can best perform a variety of tasks & applications when special-purpose equipment is not justified
- 13. Dead Weight:** Reduce the ratio of dead weight of mobile handling equipment to load carried
- 14. Utilization:** Plan for optimum utilization of handling equipment & manpower

Principles of Material Handling (Contd.)

- 15. Maintenance:** Plan for preventive maintenance & schedule repairs of all handling equipment
- 16. Obsolescence:** Replace obsolete handling methods & equipment when more efficient methods of equipment will improve operations
- 17. Control:** Use material Handling activities to improve control of production, inventory & order handling
- 18. Capacity:** Use handling equipment to help achieve the desired production capacity
- 19. Performance:** Determine the effectiveness of handling performance in terms of expense per unit handled
- 20. Safety:** Provide suitable methods & equipment for safe handling

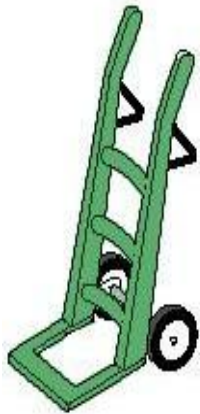
Types of Material Handling Equipment

MH EQUIPMENT TYPES

Five categories:

1. Industrial trucks
2. Automated guided vehicles
3. Monorails and other rail guided vehicles
4. Conveyors
5. Cranes and hoists

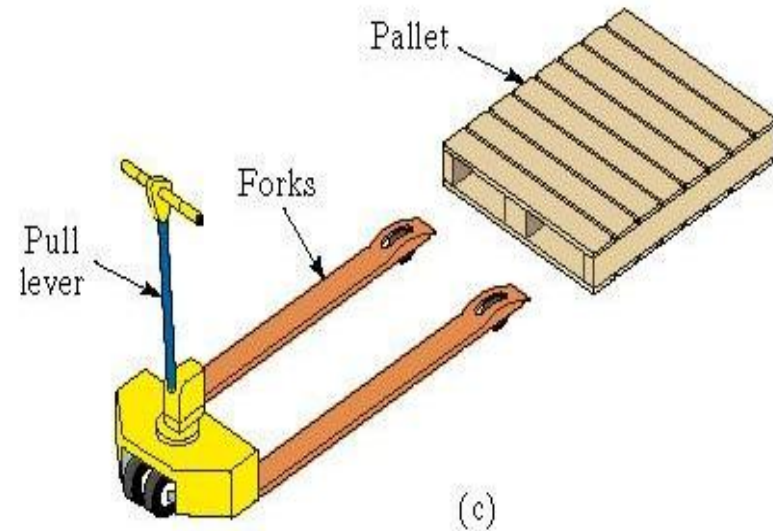
Non-powered Industrial Trucks (Hand Trucks)



(a)



(b)

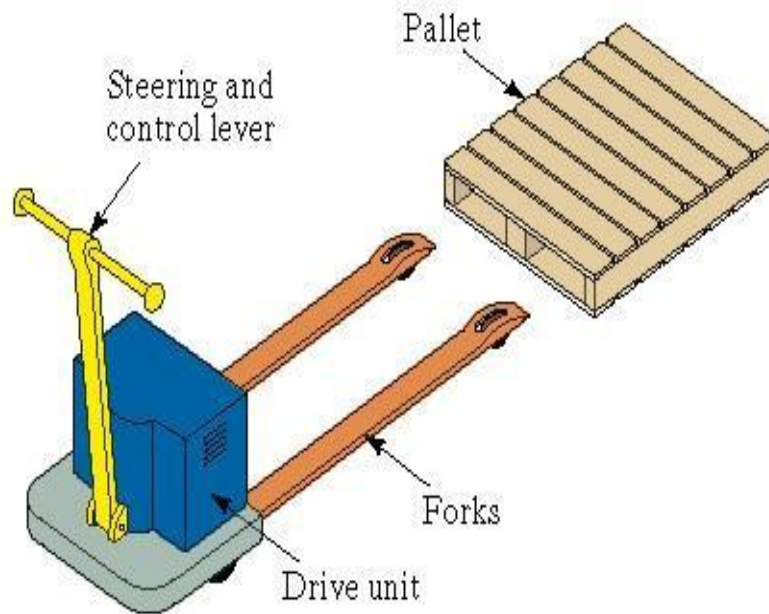


(c)

(a) Two-wheel hand truck, (b) four-wheel dolly, (c) hand-operated low-lift pallet truck

Powered Trucks

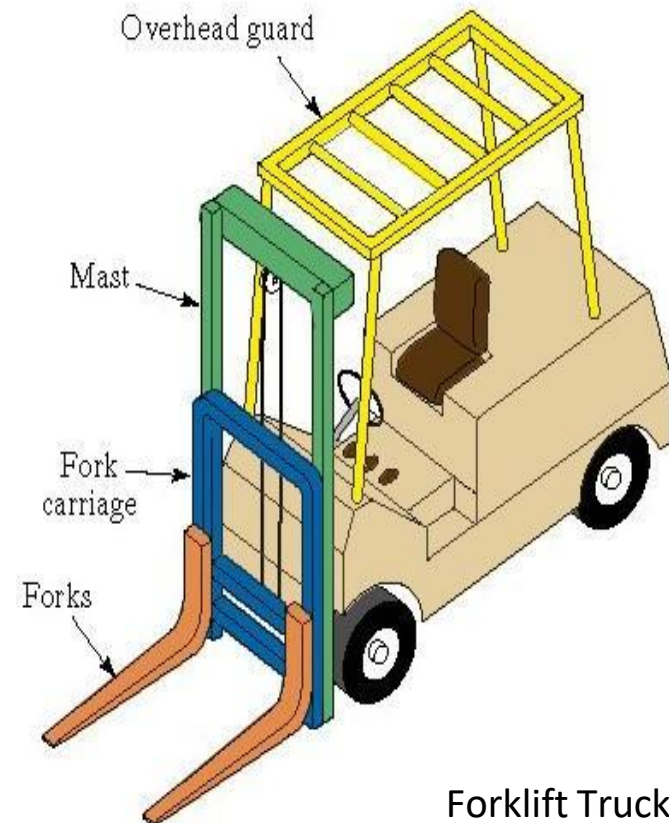
Walkie Truck



- Wheeled forks insert into pallet openings
- No provision for riding; truck is steered by worker using control handle at front of vehicle

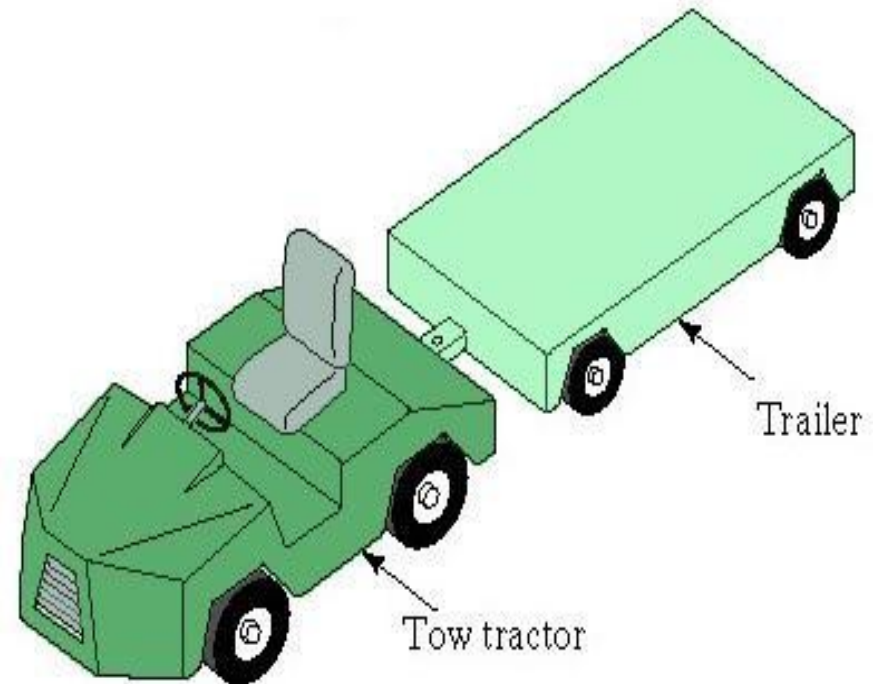
Powered Trucks

- Widely used in factories and warehouses because pallet loads are so common
- Capacities from 450 kg up to 4500 kg
- Power sources include on-board batteries and internal combustion engines



Powered Trucks

- Designed to pull one or more trailing carts in factories and warehouses, as well as for airport baggage handling
- Powered by on-board batteries or IC engines



Towing Tractor

Trailer Trains



Advantages

1. They are not require to follow a fixed path of movement & therefore can be used any where on the floor where space permits
2. They are capable of loading, unloading, & lifting, in addition to transferring material
3. Because of their unrestricted mobility, which allows them to serve different areas, trucks can achieve high utilization

Disadvantages

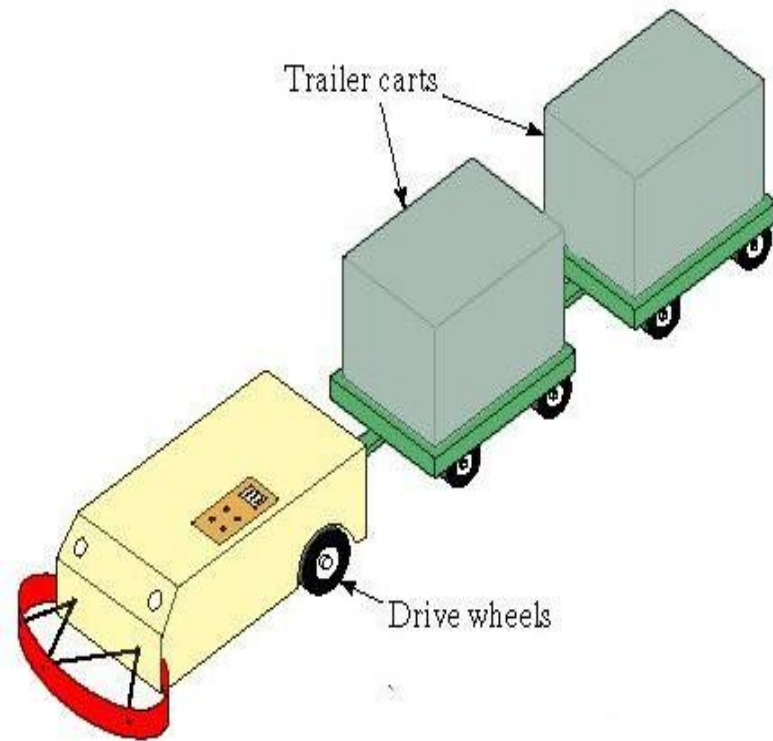
1. They cannot handle heavy loads
2. They limited capacity per trip
3. Most trucks have to be driven by an operator
4. Trucks do not allow handling to be combined with processing & inspection, as other type of equipment do

Automated Guided Vehicle System (AGVS)

- It is a material handling system that uses independently operated, self-propelled vehicles guided along defined pathways in the facility floor
- Types of AGV:
 - Driverless trains
 - Pallet trucks
 - Unit load AGVs

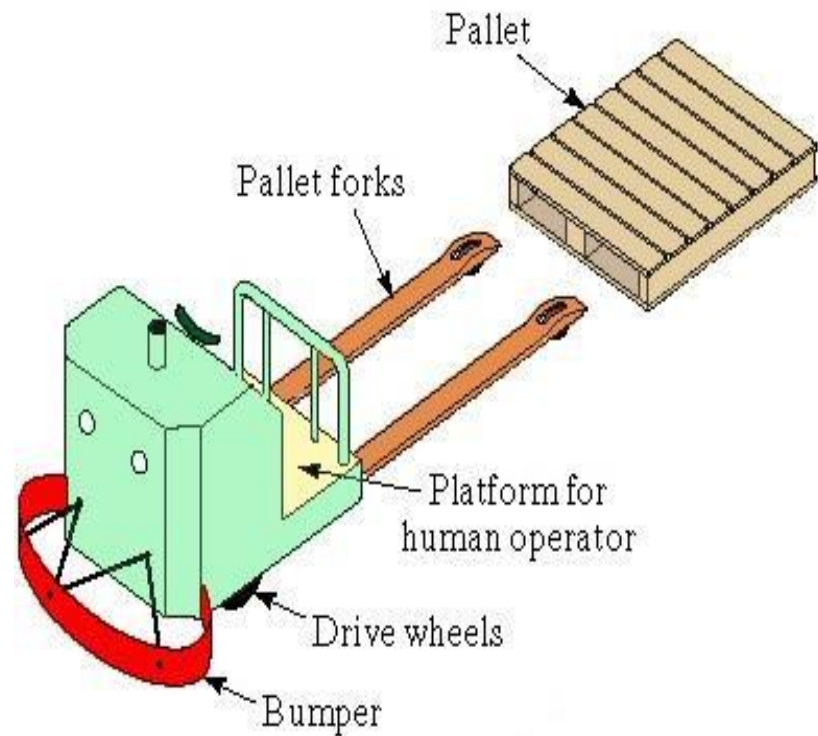
Driverless Automated Guided Train

- Common application is moving heavy payloads over long distances in warehouses and factories without intermediate stops along the route



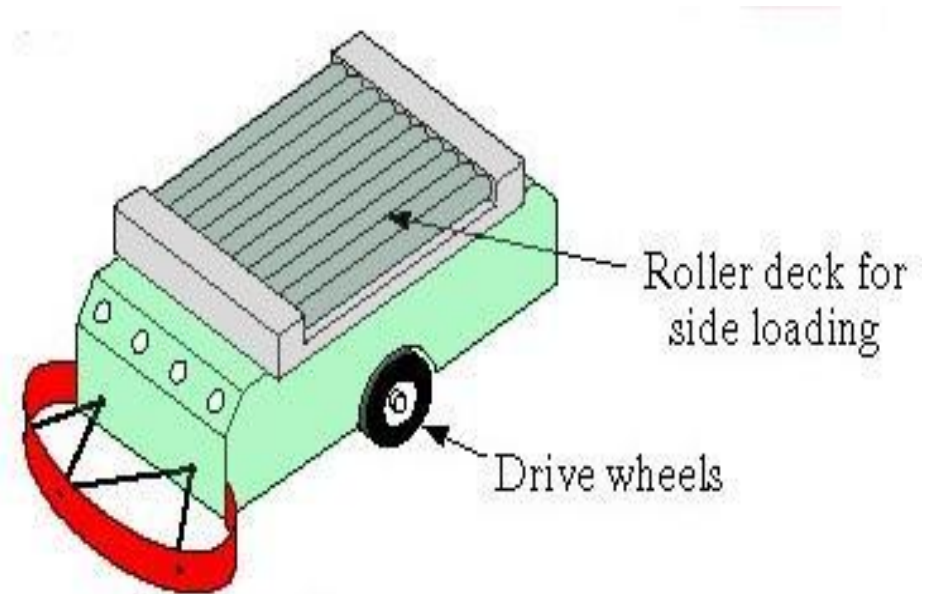
AGV Pallet Truck

- Used to move palletized loads along predetermined routes
- Vehicle is backed into loaded pallet by worker; pallet is then elevated from floor
- Worker drives pallet truck to AGV guide path and programs destination



Unit Load Carrier

- Used to move unit loads from station to station
- Often equipped for automatic loading/unloading of pallets and tote pans using roller conveyors, moving belts, or mechanized lift platforms



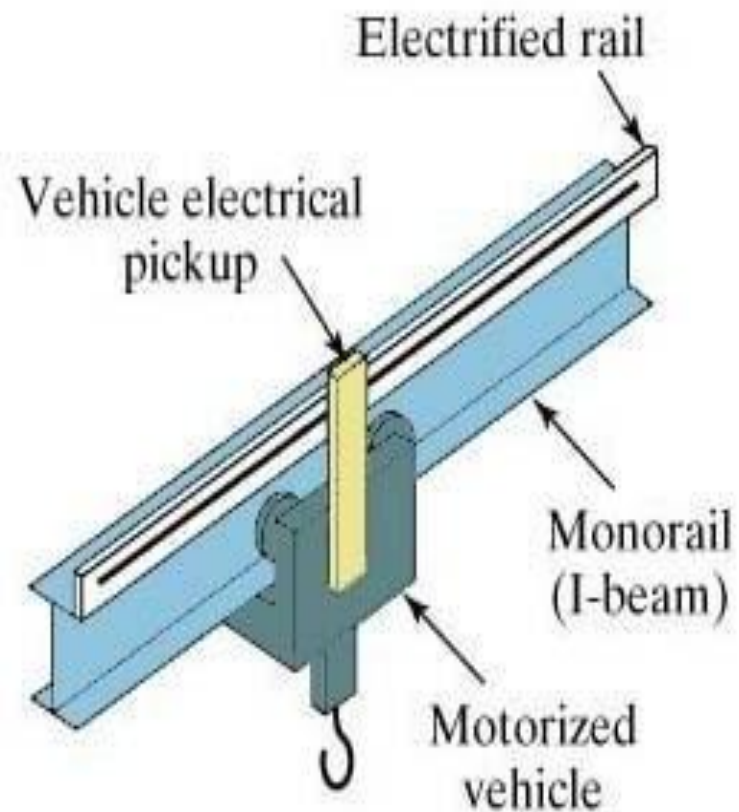
AGVs Applications

1. *Driverless train operations* - movement of large quantities of material over long distances
2. *Storage and distribution* - movement of pallet loads between shipping/receiving docks and storage racks
3. *Assembly line operations* - movement of car bodies and major subassemblies (motors) through the assembly stations
4. *Flexible manufacturing systems* - movement of work parts between machine tools
5. *Miscellaneous* - mail delivery and hospital supplies

Rail-Guided Vehicles

- Self-propelled vehicles that ride on a fixed-rail system
- Vehicles operate independently and are driven by electric motors that pick up power from an electrified rail
- Fixed rail system
 - Overhead monorail - suspended overhead from the ceiling
 - On-floor - parallel fixed rails, tracks generally protrude up from the floor
- Routing variations are possible: switches, turntables, and other special track sections

Overhead Monorail



Conveyor Systems

Large family of material transport equipment designed to move materials over fixed paths, usually in large quantities or volumes

1. Non-powered

- Materials moved by human workers or by gravity

2. Powered

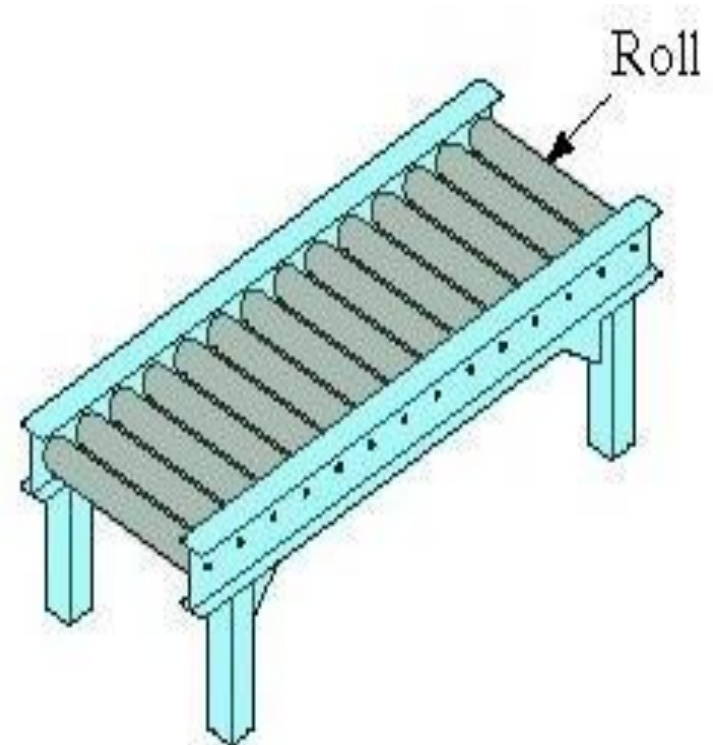
- Power mechanism for transporting materials is contained in the fixed path, using chains, belts, rollers or other mechanical devices

Conveyor Types

1. Roller
2. Skate-wheel
3. Belt
4. In-floor towline
5. Overhead trolley conveyor
6. Cart-on-track conveyor

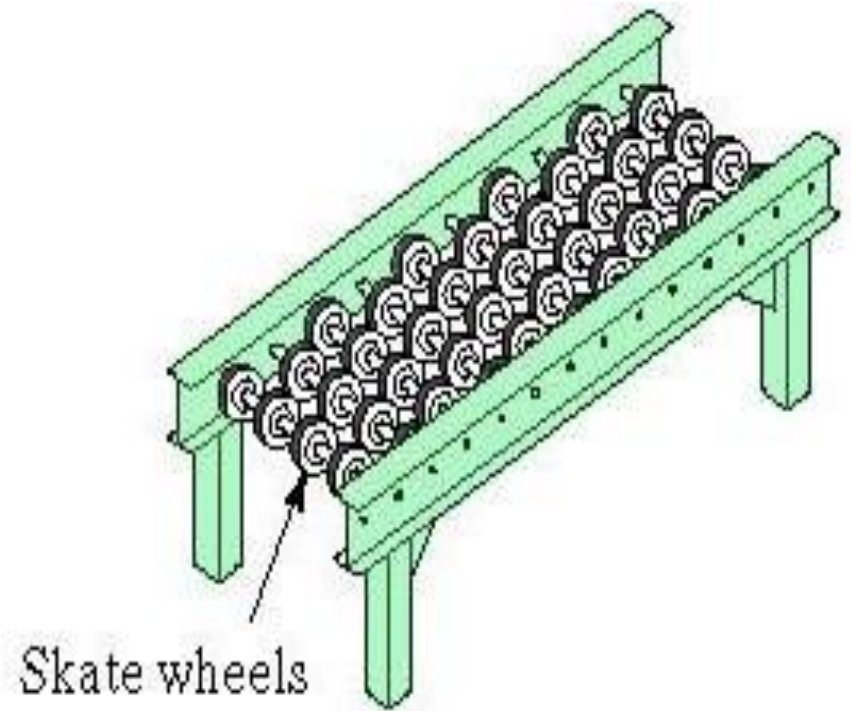
Roller Conveyor

- Pathway consists of a series of rollers that are perpendicular to direction of travel
- Loads must possess a flat bottom to span several rollers
- Powered rollers rotate to drive the loads forward
- Un-powered roller conveyors also available



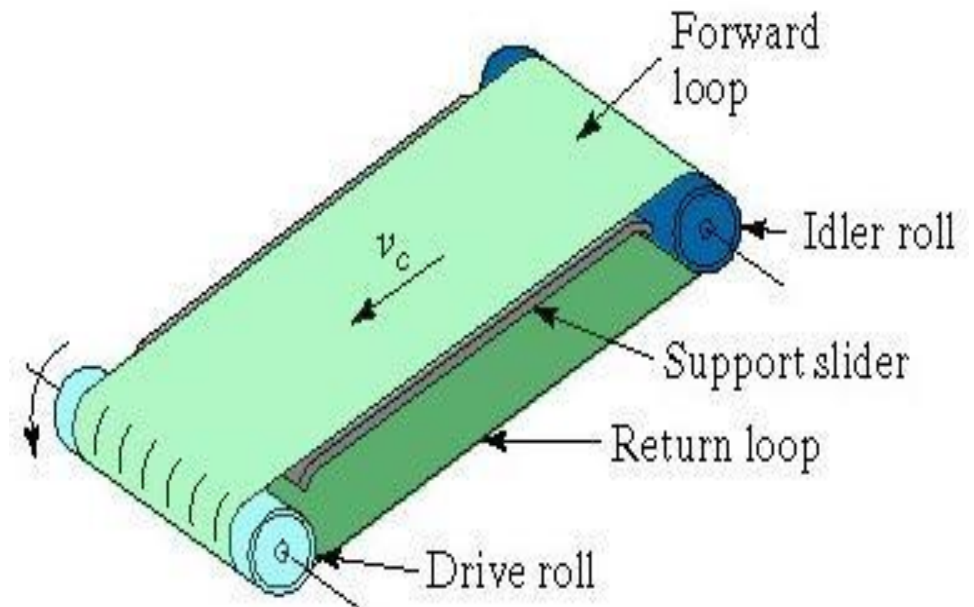
Skate-Wheel Conveyor

- Similar in operation to roller conveyor but use skate wheels instead of rollers
- Lighter weight and unpowered
- Sometimes built as portable units that can be used for loading and unloading truck trailers in shipping and receiving



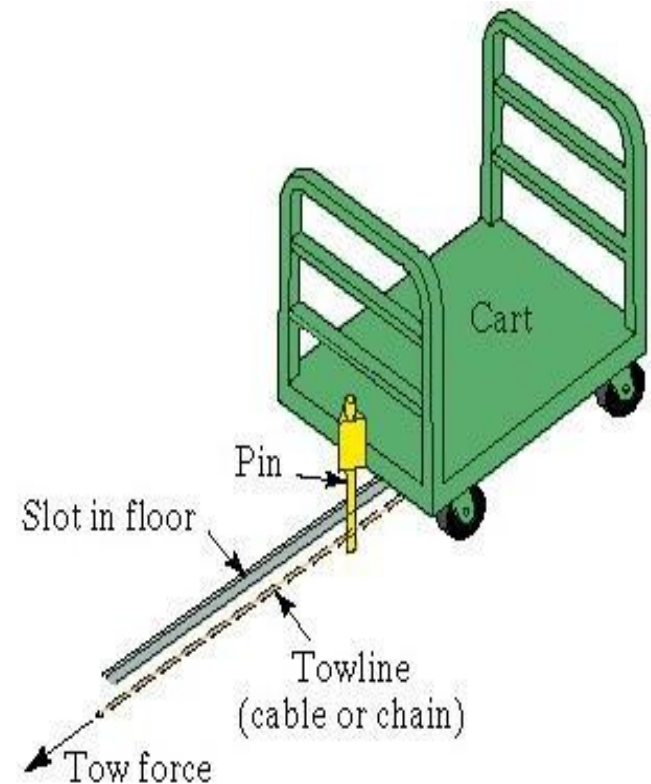
Belt Conveyor

- Continuous loop with forward path to move loads
- Belt is made of reinforced elastomer
- Support slider or rollers used to support forward loop
- Two common forms:
 - Flat belt
 - V-shaped for bulk materials



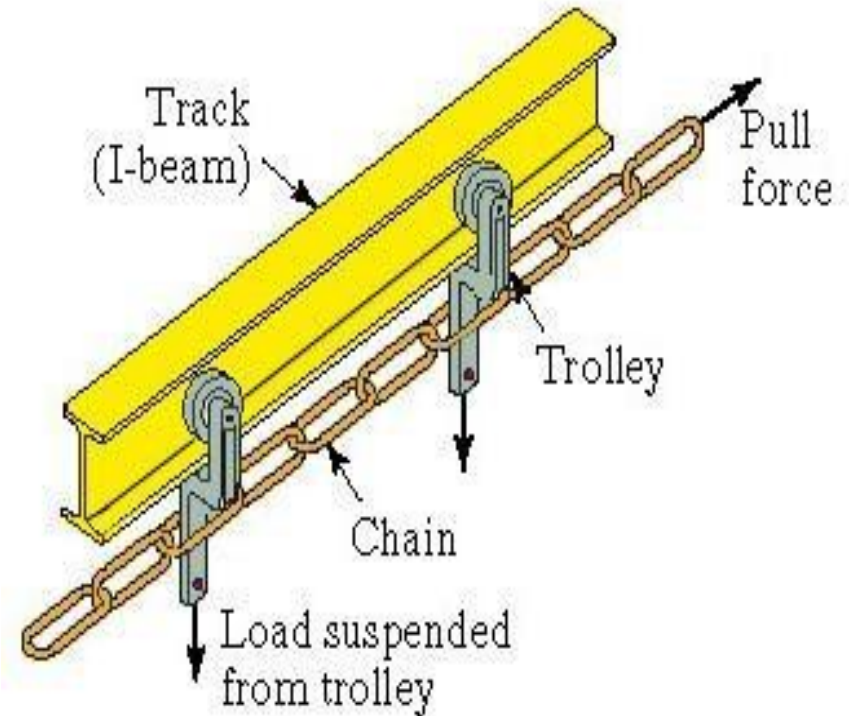
In-Floor Tow-Line Conveyor

- Four-wheel carts powered by moving chains or cables in trenches in the floor
- Carts use steel pins (or grippers) to project below floor level and engage the chain (or pulley) for towing
- This allows the carts to be disengaged from towline for loading and unloading



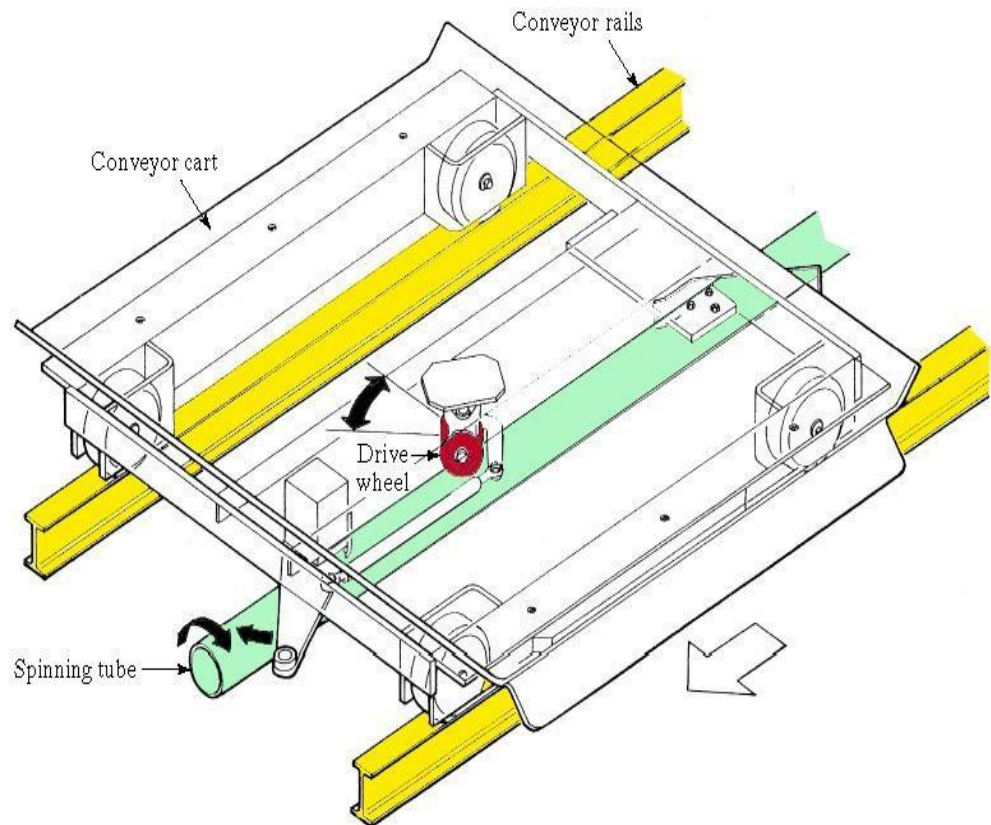
Overhead Trolley Conveyor

- A trolley is a wheeled carriage running on an overhead track from which loads can be suspended
- Trolleys are connected and moved by a chain or cable that forms a complete loop



Cart-On-Track Conveyor

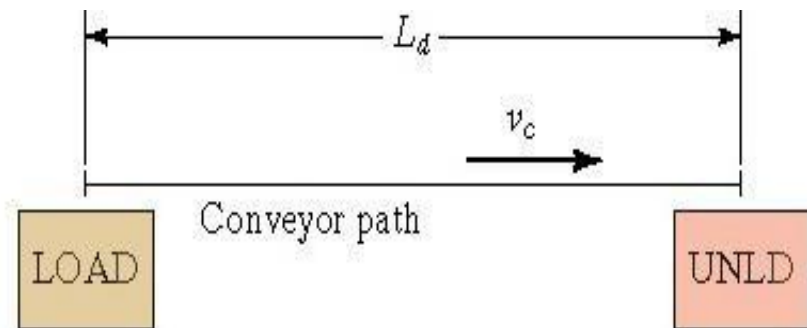
- Carts ride on a track above floor level
- Carts are driven by a spinning tube
- Forward motion of cart is controlled by a drive wheel whose angle can be changed from zero (idle) to 45 degrees (forward)



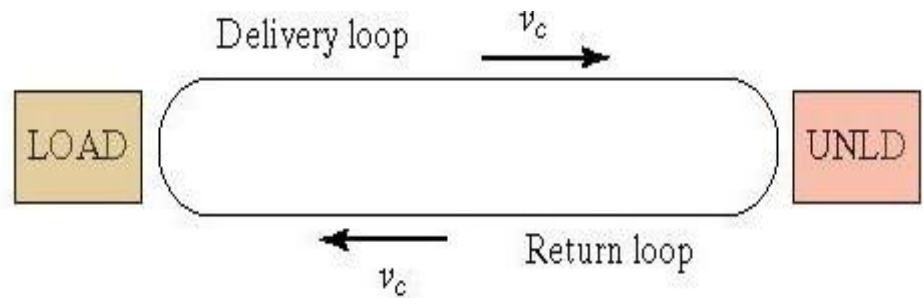
Powered Conveyor - Operations and Features

- Types of motions
 - Continuous - conveyor moves at constant velocity
 - Asynchronous - conveyor moves with stop-and-go motion
 - They stop at stations, move between stations
- Another classification of conveyors:
 - Single direction
 - Continuous loop

(a) Single direction conveyor



(b) Continuous loop conveyor



Advantages

1. Permits high capacity for moving large number of items
2. Their speed is adjustable
3. Handling combined with other activities such as processing & inspection is possible
4. They are versatile & can be on floor or overhead
5. Temporary storage of loads b/w work station is possible (particularly overhead conveyors)
6. Load transfer is automatic & does not require the assistance of many operators
7. Straight line paths or aisles are not required
8. Utilization of the cube is feasible through the use of overhead conveyors

Disadvantages:

1. They allow a fixed path, serving only limited areas
2. Bottlenecks can develop in the system
3. A breakdown in any part of the conveyor stops the entire line
4. Since conveyors are fixed in position, they hinder the movement of mobile equipment on the floor

Cranes and Hoists

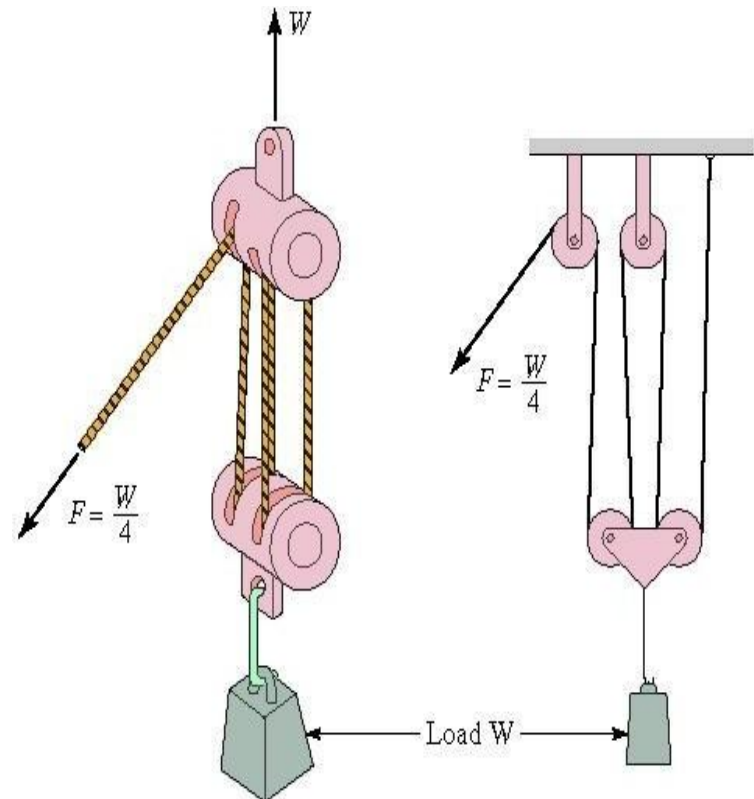
- Handling devices for lifting, lowering and transporting materials, often as heavy loads
- Cranes
 - Used for horizontal movement of materials
- Hoists
 - Used for vertical lifting of materials
- Cranes usually include hoists so that the crane-and-hoist combination provides
 - Horizontal transport
 - Vertical lifting and lowering

Hoist

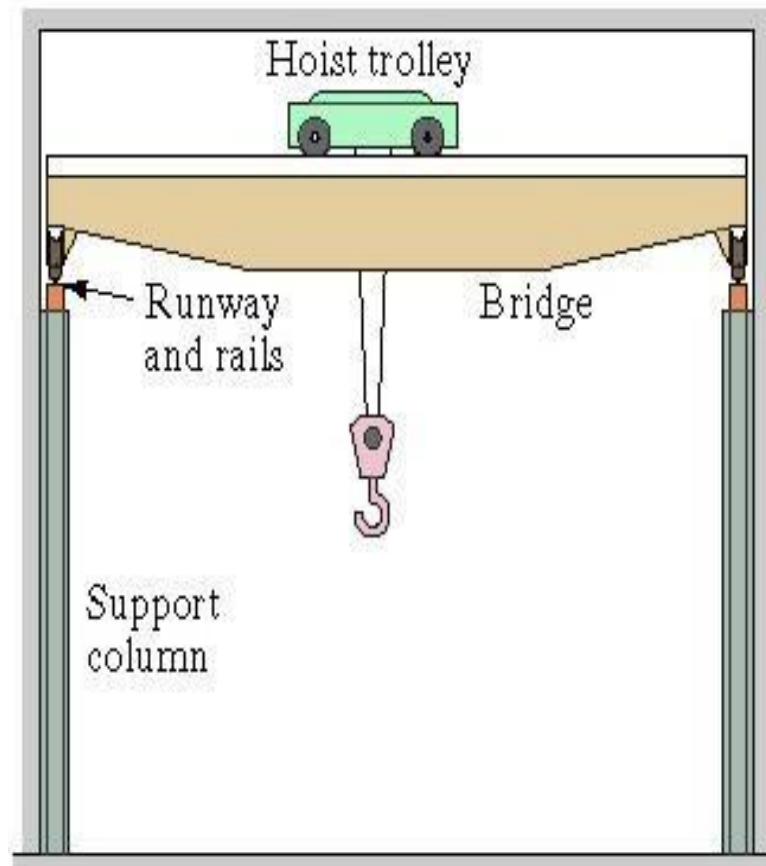
Hoist with mechanical advantage of four:

(a) sketch of the hoist

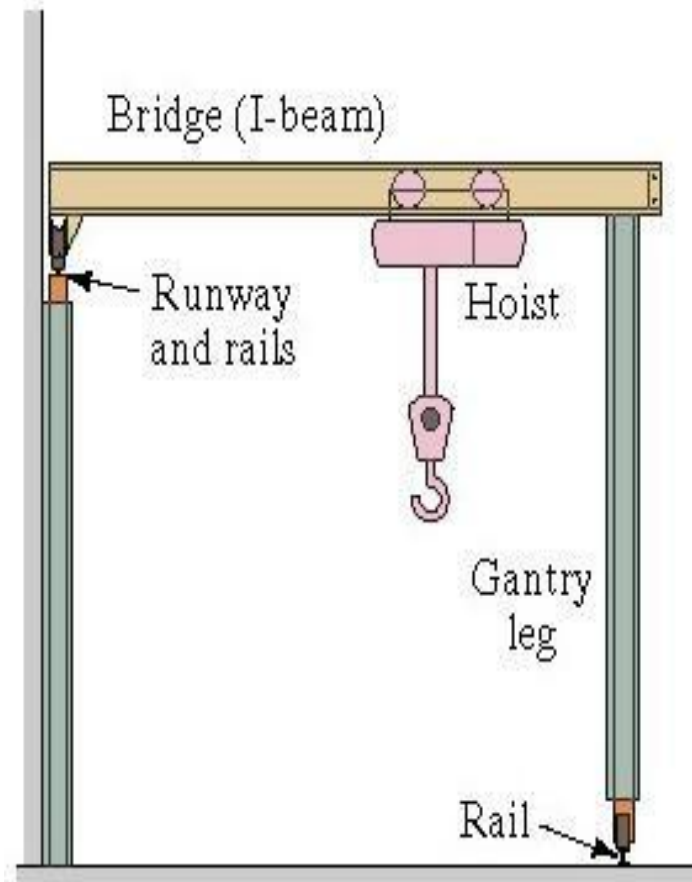
(b) diagram to illustrate mechanical advantage



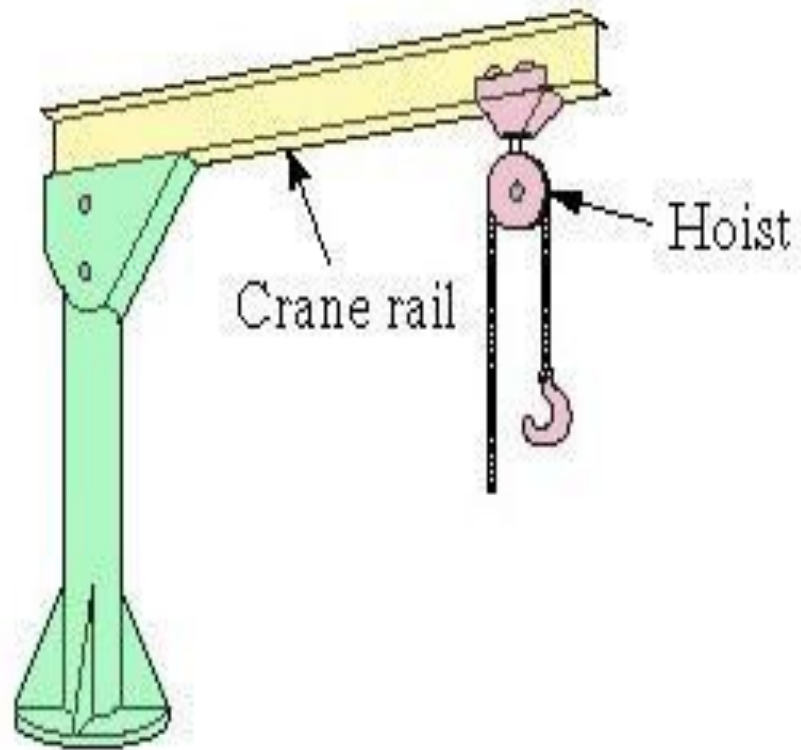
Bridge Crane



Gantry Crane



Jib Crane



Advantages:

1. Lifting as well as transferring of material is possible
2. Interference with the work on the floor is minimized
3. Valuable floor space is saved for truck rather than being utilized for installation of handling equipment
4. Such equipment is capable of handling heavy loads
5. Such equipment can be used for loading & unloading of materials

Disadvantages:

1. They require heavy investment (especially bridge cranes)
2. They serve a limited area
3. Some cranes move only in straight line & thus can not make turns
4. Utilization may not be as high as desirable since cranes are used only for a short time during daily work
5. An operator has to be available for operating some types, such as bridge cranes

Applications:

- Shipyards & heavy equipment production facilities

Degree of Mechanization in Material handling

Level of Mechanization

1. Manual & Dependent on Physical Effort

This level also includes manually driven equipment such as hand truck

2. Mechanized

Power instead of physical effort is used for driving the eqpt. Some trucks, conveyors, & cranes fall into this level. Here operators are needed for operating the eqpt as opposed providing the power

3. Mechanized Complemented with Computers

It is an extension of 2nd level

The function of the computers is to generate documents specifying the Moves & Operations

4. Automated

Minimum human intervention is used for driving & operating the eqpt, & most of these functions are performed by computers.

Examples includes Conveyors, AGVs, & AS/RS

The equipment usually receives instructions form key boards, pushbuttons, & tape or card readers

5. Fully Automated

This level is similar to the 4th level, but computers perform the additional task of on-line control, thus eliminating the need for human intervention

The cost & complexity of designing the system increases as the degree of mechanization increases. However, efficiency of operations & labour savings can result in cost & complexity justification.

References

- [http://www.mhi.org/downloads/learning/cicmhe/guidelines/10 principles.pdf](http://www.mhi.org/downloads/learning/cicmhe/guidelines/10_principles.pdf)
- [http://zfracht.ucoz.com/new 20 containers.jpg](http://zfracht.ucoz.com/new_20_containers.jpg)